

# Synthetic Fixed-Confidence Experiments for Binary Pairwise Pareto Set Identification

Experiment report generated from `Exp_Synthetic`

July 8, 2026

## Abstract

This report summarizes the completed synthetic experiments for fixed-confidence binary pairwise Pareto set identification. The experiments compare VB-EGE-practical against three fixed-confidence benchmarks: UniformFocalBorda-FC, UniformPairwiseBT-MLE-FC, and UniformPairwiseBT-BordaPlugIn-FC. The main runs contain 3400 scaling replications and 2700 benchmark replications per algorithm across all experiment cells. Across all completed main settings, the empirical error rate is 0.0 for scaling and 0.0 for benchmarks; therefore the main comparison is sample complexity, measured by mean stopping time. The experiments are organized in logical order: fixed-confidence scaling, confidence-scaling quantiles, the fixed-confidence benchmark baseline suite, constant calibration, Pareto-size ablation, correlated arena instances, pair-cell coverage diagnostics, fixed-error budget calibration, and zero-error diagnostics.

## 1 Protocol

The task is fixed-confidence identification of the strict Pareto set under binary pairwise comparison feedback. Each run stops adaptively when the algorithm can return a Pareto set certificate at the target failure probability. The default target is  $\delta = 0.05$ , except in the confidence-scaling sweep where  $\delta$  itself is varied. The reported stopping time counts coordinate-specific pairwise comparison pulls. Lower stopping time is better when the empirical error rate is controlled.

Table 1: Algorithms compared in the fixed-confidence experiments.

| Name                             | Sampling model                                          | Fixed-confidence rule                                             |
|----------------------------------|---------------------------------------------------------|-------------------------------------------------------------------|
| VB-EGE-practical                 | Coordinate-wise Vector-Borda estimates                  | Adaptive accept/reject using empirical gap certificates           |
| UniformFocalBorda-FC             | Uniform coordinate-wise Borda sampling                  | Same confidence threshold family without adaptive elimination     |
| UniformPairwiseBT-MLE-FC         | Uniform pair-coordinate Bradley–Terry sampling with MLE | Stop when plug-in Pareto set is certified by BT confidence radius |
| UniformPairwiseBT-BordaPlugIn-FC | Uniform pair-coordinate sampling with Borda plug-in     | Stop when Borda plug-in certificate is satisfied                  |

**Definitions used by the report.** There are  $K$  arms and  $d$  objectives. A coordinate-specific comparison between arms  $i$  and  $j$  on objective  $r$  follows a Bradley–Terry model,

$$Y_{ijr} \sim \text{Bernoulli}(p_{ijr}), \quad p_{ijr} = \sigma(\theta_{ir} - \theta_{jr}).$$

The coordinate-wise Borda embedding is

$$B_{ir} = \frac{1}{K-1} \sum_{j \neq i} p_{ijr}.$$

Most certificates in this report are computed in this Vector-Borda space. The synthetic generators used for the main experiments pass a Borda/latent Pareto consistency check, so the reported true Pareto set agrees under  $\theta$  and under  $B$ .

For any score matrix  $X \in \mathbb{R}^{K \times d}$ , define

$$m_{ij}(X) = \min_r (X_{jr} - X_{ir}), \quad M_{ij}(X) = \max_r (X_{ir} - X_{jr}), \quad \Delta_i^*(X) = \max_{j \neq i} m_{ij}(X).$$

If arm  $i$  is non-Pareto, its identification gap is  $\Delta_i(X) = \Delta_i^*(X)$ . If arm  $i$  is Pareto, the reported gap is

$$\Delta_i(X) = \min_{j \neq i} \min \{M_{ij}(X), \max(M_{ji}(X), 0) + \max(\Delta_j^*(X), 0)\}.$$

The minimum Borda gap is  $\Delta_{\min, B} = \min_i \Delta_i(B)$ . The Borda hardness is

$$H_B = \sum_i \Delta_i(B)^{-2},$$

with  $H_B = \infty$  when a gap is non-positive or non-finite. Large  $H_B$  means that one or more arms lie close to the Pareto decision boundary; many plots therefore also report  $\tau/(dH_B)$  to remove first-order instance hardness.

All fixed-confidence methods use the same practical constants:  $c_{\text{samp}} = 2$ ,  $c_{\text{thr}} = 4$ , and  $c_{\text{log}} = 4$ . In phase  $m$ , with cell count  $C$ , the schedule is

$$\varepsilon_m = 2^{-m}, \quad L_m = \log(c_{\text{log}} C m^2 / \delta), \quad n_m = \lceil c_{\text{samp}} L_m / \varepsilon_m^2 \rceil, \quad r_m = \sqrt{L_m / (2n_m)}.$$

The sample constant  $c_{\text{samp}}$  multiplies the per-cell sample count, the log constant  $c_{\text{log}}$  is the union-bound safety factor inside  $L_m$ , and the threshold constant  $c_{\text{thr}}$  is the certificate margin. A fixed-confidence baseline stops when its estimated minimum gap satisfies  $\hat{\Delta}_{\min} > c_{\text{thr}} r_m$ .

**Benchmark implementations.** The three benchmarks other than VB-EGE are implemented as follows.

- **UniformFocalBorda-FC.** In each phase it samples every arm-objective cell  $(i, r)$  up to  $n_m$  Borda observations. Equivalently, the simulation draws  $S_{ir} \sim \text{Binomial}(N_{ir}, B_{ir})$  and estimates  $\hat{B}_{ir} = S_{ir}/N_{ir}$ . It computes the plug-in strict Pareto set from  $\hat{B}$ , computes  $\hat{\Delta}_{\min}$  using the gap formula above, and stops when  $\hat{\Delta}_{\min} > c_{\text{thr}} r_m$ . It does not eliminate arms adaptively; it keeps all  $Kd$  cells balanced.
- **UniformPairwiseBT-MLE-FC.** In each phase it samples every pair-coordinate cell  $(i, j, r)$ ,  $i < j$ , up to  $n_m$  comparisons, so  $C = dK(K-1)/2$ . For each objective  $r$ , it fits a separate ridge-stabilized BT MLE from the win counts:

$$\min_{\vartheta_{\cdot r}} \sum_{i < j} \{W_{ijr} \log(1 + \exp[-(\vartheta_{ir} - \vartheta_{jr})]) + W_{jir} \log(1 + \exp[\vartheta_{ir} - \vartheta_{jr}])\} + \frac{\lambda}{2} \|\vartheta_{\cdot r}\|_2^2.$$

The fitted coordinate scores are centered for identifiability. The algorithm then computes the plug-in Pareto set and  $\hat{\Delta}_{\min}$  from  $\hat{\theta}$  and stops when the same gap certificate is satisfied. If the graph is disconnected, the optimizer fails, or the estimate is too large, the implementation reruns the fit with the fallback ridge value.

- **UniformPairwiseBT-BordaPlugIn-FC.** It uses the same balanced pair-coordinate sampling as the MLE baseline, but avoids solving the BT MLE. It estimates each pairwise probability with add-1/2 smoothing,

$$\hat{p}_{ijr} = \frac{W_{ijr} + 1/2}{N_{ijr} + 1},$$

forms the plug-in Borda estimate  $\hat{B}_{ir} = (K - 1)^{-1} \sum_{j \neq i} \hat{p}_{ijr}$ , and applies the same strict-Pareto plug-in set and gap certificate in Borda space.

In the benchmark runs, the BT-MLE baseline uses balanced random pair-coordinate allocation, ridge parameter  $10^{-8}$ , fallback ridge parameter  $10^{-4}$ , maximum absolute latent score 20, and MLE tolerance  $10^{-9}$  in the benchmark run. Raw outputs are stored as CSV files because the local parquet engine was unavailable; this does not change the numerical results.

**Metrics and legends.** The main metric is mean stopping time  $\bar{\tau}$ . The normalized budget  $\tau/(dH_B)$  uses the instance Borda hardness  $H_B$  and is used only to put settings with different dimensions and gaps on a comparable axis. Across the main fixed-confidence runs the empirical Pareto-set error is zero, so the all-zero error floor plots are reported as zero-error diagnostics rather than as the primary performance metric. The formal baselines in the report are the fixed-confidence baselines, not capped fixed-budget variants. The fixed-error budget calibration uses capped variants only to show what happens before certification. In line plots, the legend entries are the four algorithm names. In bar plots, the x-axis labels are algorithm names and the y-axis is log-scaled mean stopping time. In pair-cell coverage plots, the vertical dashed line at coverage 1 means one sample per pair-coordinate cell on average.

**Plot computation conventions.** Whenever the same axis label appears later, it uses the same computation. Mean stopping time is  $\bar{\tau} = n^{-1} \sum_s \tau_s$  over replications. Median, q90, and q95 stopping times are empirical quantiles of the same  $\{\tau_s\}$  values. Empirical error is  $n^{-1} \sum_s \mathbf{1}\{\hat{P}_s \neq P_s\}$ ; when no failures are observed, error-floor plots show  $\log_{10}(0.5/n)$  instead of  $-\infty$ . The Wilson upper bound is the upper endpoint of the two-sided 95% Wilson interval for this binomial error rate. A baseline/VB ratio is  $\bar{\tau}_{\text{baseline}}/\bar{\tau}_{\text{VB}}$  within the same setting. Pair-cell coverage is averaged over runs after computing  $\tau_s/C$  for each pairwise run, where  $C = dK(K - 1)/2$ . Mean Hamming distance is the average symmetric-difference size  $|\hat{P}_s \Delta P_s|$ . The fixed-error budget ratio is  $T/\bar{\tau}_{\text{VB,ref}}$ , where  $\bar{\tau}_{\text{VB,ref}}$  is the default fixed-confidence VB-EGE mean stopping time on the matching benchmark setting. Final phase is the last dyadic phase index  $m$  used by the algorithm. Achieved objective correlation is the mean off-diagonal sample correlation among objective columns of the generated instance.

## 2 Fixed-Confidence Scaling

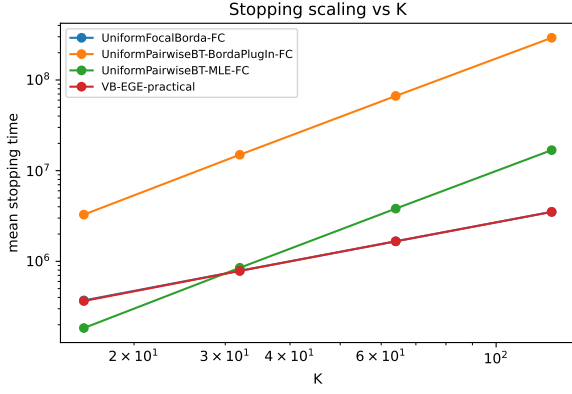
Table 2: Fixed-confidence scaling settings. All tasks use the symmetric hard generator and four algorithms; each cell is averaged over 200 random permutations/seeds.

| Sweep              | Generator      | Grid                                                     | Purpose                                                                            |
|--------------------|----------------|----------------------------------------------------------|------------------------------------------------------------------------------------|
| K scaling          | symmetric_hard | K in {16,32,64,128}; d=4; Delta=1; delta=0.05            | One Pareto arm at the origin and K-1 dominated arms at -Delta in every coordinate. |
| d scaling          | symmetric_hard | K=64; d in {2,4,10}; Delta=1; delta=0.05                 | Symmetric hard instance with the number of dimensions varied.                      |
| Gap scaling        | symmetric_hard | K=64; d=10; Delta in {0.5,0.75,1,1.25,1.5}; delta=0.05   | Symmetric hard instance with the latent separation Delta varied.                   |
| Confidence scaling | symmetric_hard | K=64; d=10; Delta=1; delta in {0.20,0.10,0.05,0.02,0.01} | Symmetric hard instance with the target failure probability varied.                |

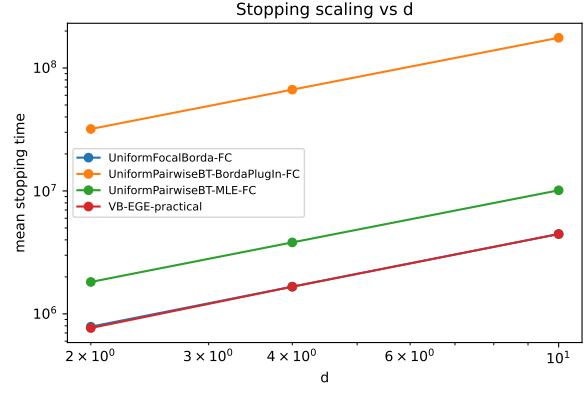
Table 3: Scaling results. For each swept parameter value, the table reports VB-EGE mean stopping time and baseline ratios relative to VB-EGE. All empirical error rates are zero.

| Sweep              | Value | VB-EGE $\bar{\tau}$ | Focal/VB | MLE/VB | Plug-in/VB |
|--------------------|-------|---------------------|----------|--------|------------|
| K scaling          | 16    | $3.65 \times 10^5$  | 1.02x    | 0.50x  | 8.97x      |
| K scaling          | 32    | $7.84 \times 10^5$  | 1.00x    | 1.08x  | 19.1x      |
| K scaling          | 64    | $1.66 \times 10^6$  | 1.00x    | 2.29x  | 40.1x      |
| K scaling          | 128   | $3.51 \times 10^6$  | 1.00x    | 4.80x  | 83.2x      |
| d scaling          | 2     | $7.67 \times 10^5$  | 1.03x    | 2.37x  | 41.6x      |
| d scaling          | 4     | $1.66 \times 10^6$  | 1.00x    | 2.29x  | 40.1x      |
| d scaling          | 10    | $4.46 \times 10^6$  | 1.00x    | 2.27x  | 39.5x      |
| Gap scaling        | 0.50  | $1.84 \times 10^7$  | 1.00x    | 2.31x  | 39.2x      |
| Gap scaling        | 0.75  | $4.46 \times 10^6$  | 1.00x    | 2.27x  | 39.5x      |
| Gap scaling        | 1     | $4.46 \times 10^6$  | 1.00x    | 2.27x  | 39.5x      |
| Gap scaling        | 1.25  | $1.82 \times 10^6$  | 2.44x    | 1.27x  | 23.4x      |
| Gap scaling        | 1.50  | $1.07 \times 10^6$  | 1.00x    | 2.17x  | 39.8x      |
| Confidence scaling | 0.01  | $4.99 \times 10^6$  | 1.00x    | 2.24x  | 38.6x      |
| Confidence scaling | 0.02  | $4.76 \times 10^6$  | 1.00x    | 2.25x  | 39.0x      |
| Confidence scaling | 0.05  | $4.46 \times 10^6$  | 1.00x    | 2.27x  | 39.5x      |
| Confidence scaling | 0.10  | $4.23 \times 10^6$  | 1.00x    | 2.29x  | 39.9x      |
| Confidence scaling | 0.20  | $4.01 \times 10^6$  | 1.00x    | 2.30x  | 40.4x      |

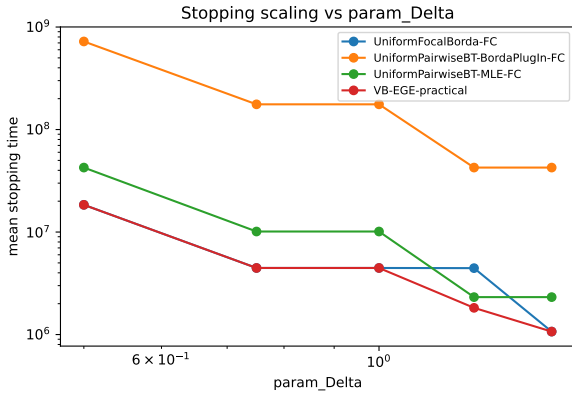
The scaling tasks isolate how stopping time changes with the number of arms, the dimension, the latent gap, and the confidence level on the symmetric hard instance. In this generator, there is a single true Pareto arm and all other arms are dominated by the same coordinate gap. This makes the setting useful for checking whether the fixed-confidence rules behave monotonically and whether pairwise baselines pay a large cost for broad pair-coordinate coverage.



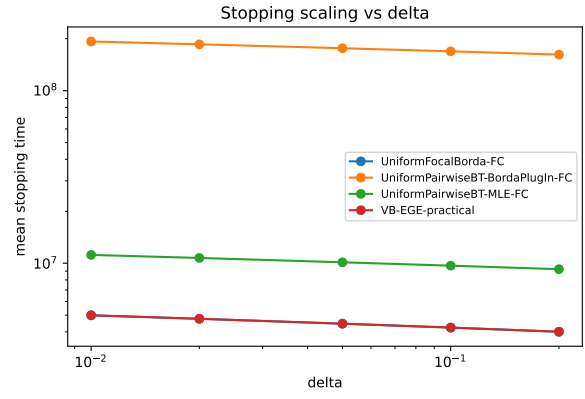
(a) K scaling



(b) d scaling



(c) Gap scaling



(d) Confidence scaling

Figure 1: Scaling curves for mean stopping time. In each panel, the x-axis is the swept design parameter shown in the subcaption; the y-axis is mean stopping time  $\bar{\tau}$  over replications. Both axes are log-scaled when positive, and lower curves indicate fewer comparison pulls.

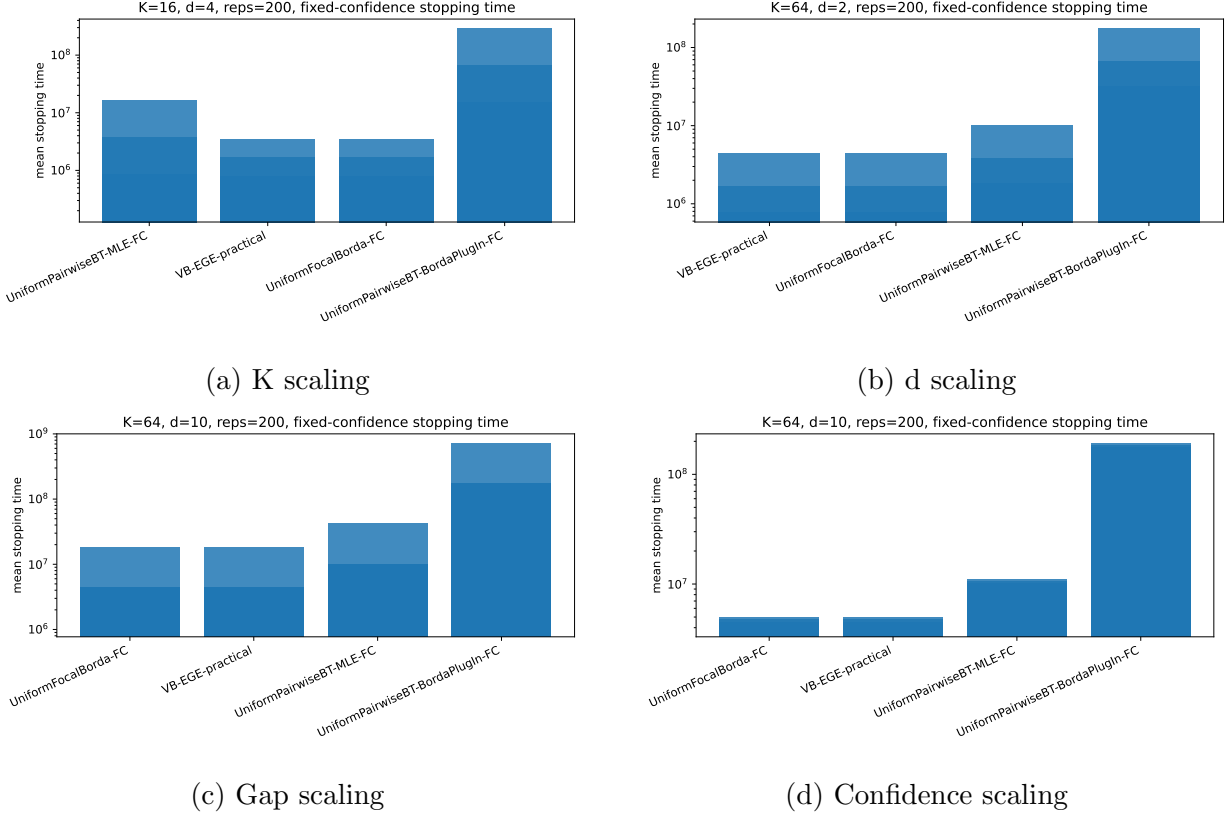


Figure 2: Per-sweep fixed-confidence stopping bars. The x-axis lists algorithms within each setting; the y-axis is mean stopping time  $\bar{\tau}$  on a log scale. Bars are sorted by mean stopping time within each plotted setting.

Result interpretation. VB-EGE-practical and UniformFocalBorda-FC are nearly tied on the symmetric hard instance, as expected because the instance has a single symmetric Pareto arm and little need for adaptive frontier structure. Both methods estimate the same coordinate-wise Vector-Borda cells, and the symmetric generator gives all dominated arms comparable gaps. Therefore most certificates mature in the same final phase, leaving little heterogeneity for VB-EGE’s adaptive elimination to exploit. In this setting the near-overlap between the UniformFocalBorda-FC line and the VB-EGE-practical line is a sanity-check outcome rather than evidence against VB-EGE; the method’s advantage appears when the instance has richer frontier structure or when pairwise baselines must pay broad pair-coordinate coverage costs. The pairwise BT-MLE and Borda plug-in baselines are substantially slower as  $K$  and  $d$  grow, reflecting the cost of spreading samples across many pair-coordinate cells. Larger gaps reduce stopping time, while smaller target delta increases stopping time through the confidence threshold. The small plateaus in the gap sweep are consistent with dyadic phase schedules: once two gap values trigger the same final phase, their stopping times can match even if the underlying gap differs.

### 3 Confidence-Scaling Quantiles

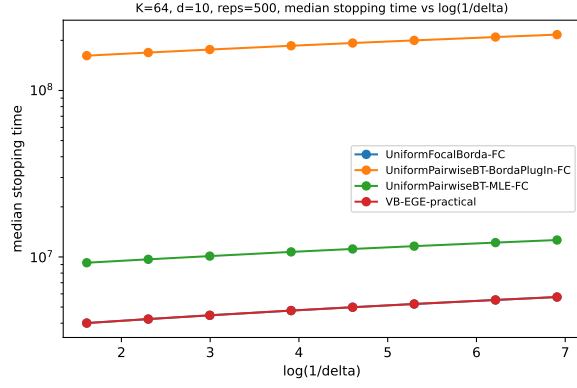
Table 4: Confidence-scaling quantile settings. These runs use the same fixed-confidence algorithms as the main benchmark suite and vary only the requested failure probability.

| Setting           | Generator               | Grid                                                                             | Purpose                                                                       | Reps |
|-------------------|-------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------|
| Symmetric K64 d10 | symmetric_hard          | K=64; d=10; Delta=1; delta in {0.20,0.10,0.05,0.02,0.01,0.005,0.002,0.001}       | Stress the tail of fixed-confidence stopping times as $\log(1/\delta)$ grows. | 500  |
| Arena-10 medium   | arena_tradeoff_frontier | K=64; d=10; s=16; margins [0.06,0.20]; delta in {0.20,0.10,0.05,0.02,0.01,0.005} | Check confidence scaling on a high-dimensional multi-Pareto arena.            | 300  |

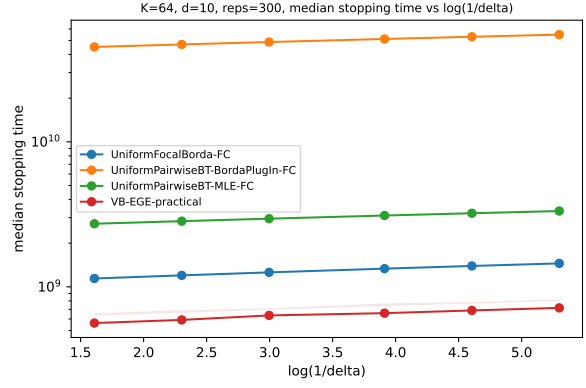
Table 5: Confidence-scaling quantile results. VB columns report mean and 95th-percentile stopping times; ratios use mean stopping time relative to VB-EGE.

| Setting           | $\delta$ | VB mean $\tau$     | VB q95 $\tau$      | Focal/VB | MLE/VB | Plug-in/VB |
|-------------------|----------|--------------------|--------------------|----------|--------|------------|
| Symmetric K64 d10 | 0.200    | $4.01 \times 10^6$ | $4.01 \times 10^6$ | 1.00x    | 2.30x  | 40.4x      |
| Symmetric K64 d10 | 0.100    | $4.23 \times 10^6$ | $4.23 \times 10^6$ | 1.00x    | 2.29x  | 39.9x      |
| Symmetric K64 d10 | 0.050    | $4.46 \times 10^6$ | $4.46 \times 10^6$ | 1.00x    | 2.27x  | 39.5x      |
| Symmetric K64 d10 | 0.020    | $4.76 \times 10^6$ | $4.76 \times 10^6$ | 1.00x    | 2.25x  | 39.0x      |
| Symmetric K64 d10 | 0.010    | $4.99 \times 10^6$ | $4.99 \times 10^6$ | 1.00x    | 2.24x  | 38.6x      |
| Symmetric K64 d10 | 0.005    | $5.22 \times 10^6$ | $5.22 \times 10^6$ | 1.00x    | 2.23x  | 38.3x      |
| Symmetric K64 d10 | 0.002    | $5.52 \times 10^6$ | $5.52 \times 10^6$ | 1.00x    | 2.21x  | 38.0x      |
| Symmetric K64 d10 | 0.001    | $5.74 \times 10^6$ | $5.74 \times 10^6$ | 1.00x    | 2.20x  | 37.7x      |
| Arena-10 medium   | 0.200    | $5.62 \times 10^8$ | $6.57 \times 10^8$ | 2.09x    | 4.99x  | 82.7x      |
| Arena-10 medium   | 0.100    | $5.89 \times 10^8$ | $6.91 \times 10^8$ | 2.10x    | 5.01x  | 81.8x      |
| Arena-10 medium   | 0.050    | $6.21 \times 10^8$ | $7.10 \times 10^8$ | 2.09x    | 4.95x  | 80.9x      |
| Arena-10 medium   | 0.020    | $6.49 \times 10^8$ | $7.69 \times 10^8$ | 2.12x    | 4.93x  | 81.1x      |
| Arena-10 medium   | 0.010    | $6.81 \times 10^8$ | $7.86 \times 10^8$ | 2.13x    | 4.92x  | 80.2x      |
| Arena-10 medium   | 0.005    | $7.06 \times 10^8$ | $8.20 \times 10^8$ | 2.12x    | 4.87x  | 79.9x      |

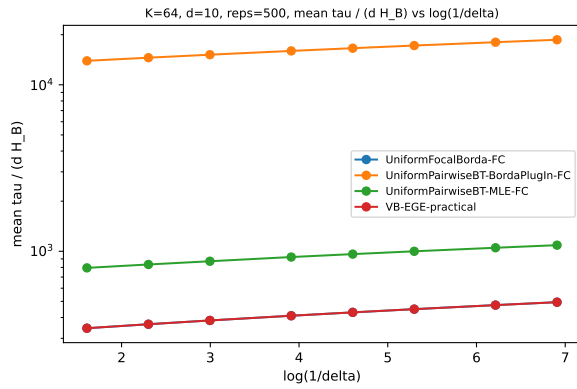
This experiment repeats fixed-confidence runs over a wider grid of target failure probabilities. Within each experiment and replication, all  $\delta$  values reuse the same synthetic instance, so the confidence effect is not confounded with between-instance hardness variation. The line legends are the four algorithms; the quantile traces report the median and upper stopping-time quantiles. The x-axis is  $\log(1/\delta)$ , so approximately linear growth indicates the expected logarithmic confidence cost. The normalized panels divide by the instance Borda hardness, and the  $\tau/\log(1/\delta)$  panels make deviations from pure logarithmic scaling visible.



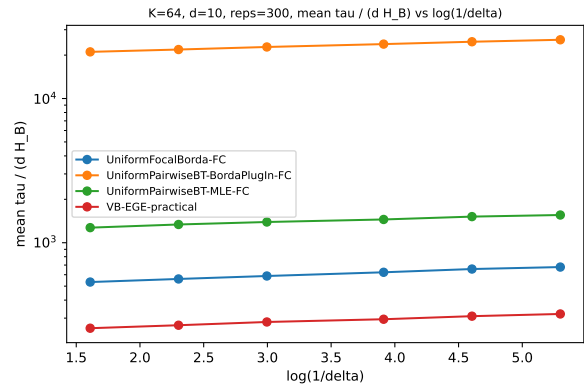
(a) Symmetric K64 d10



(b) Arena-10 medium



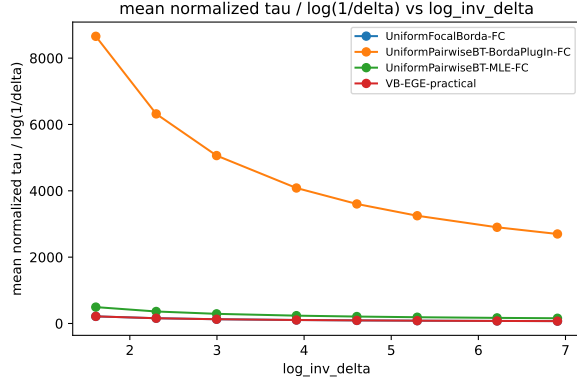
(c) Normalized symmetric



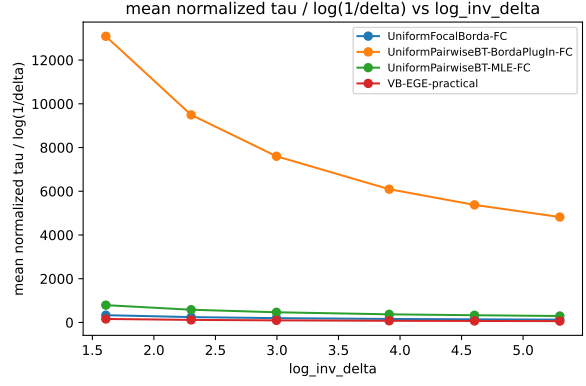
(d) Normalized arena10

Figure 3: Confidence-scaling stopping quantiles. The x-axis is  $\log(1/\delta)$ . Non-normalized panels use median  $\tau$  on the y-axis with q90–q95 shaded bands; normalized panels use mean  $\tau/(dH_B)$ . Each color is one algorithm.

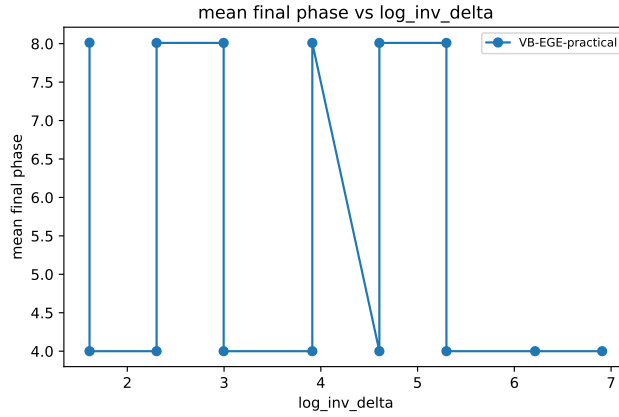




(a) Symmetric tau/log



(b) Arena10 tau/log



(c) VB final phase

Figure 4: Confidence-scaling diagnostics. The tau/log panels plot mean  $\tau/\log(1/\delta)$  against  $\log(1/\delta)$ . The final-phase panel plots mean final phase index  $m$  against  $\delta$  for VB-EGE.

Result interpretation. On the symmetric setting, VB-EGE and the focal Borda baseline remain close because the certificate is essentially one-dimensional after symmetry. The arena setting separates the methods more clearly: VB-EGE keeps a smaller high quantile stopping time than the pairwise baselines because it does not need broad pair-coordinate coverage at every confidence level. The final-phase diagnostic should move in steps rather than smoothly, which is expected from phase doubling.

## 4 Fixed-Confidence Benchmark Suite

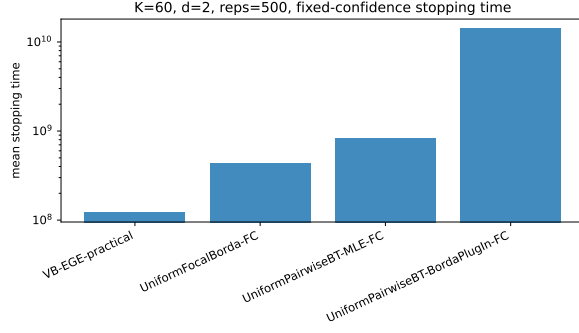
Table 6: Fixed-confidence benchmark settings. All algorithms use target failure probability  $\delta = 0.05$ ;  $|P|$  denotes the true Pareto set size.

| Setting         | Generator               | $K$ | $d$ | $ P $ | Reps | Parameters                           |
|-----------------|-------------------------|-----|-----|-------|------|--------------------------------------|
| Convex-2D       | convex_frontier_2d      | 60  | 2   | 15    | 500  | s=15                                 |
| Arena-4 small   | arena_tradeoff_frontier | 32  | 4   | 8     | 500  | s=8, margins [0.08,0.25], alpha=0.7  |
| Arena-4 medium  | arena_tradeoff_frontier | 64  | 4   | 12    | 300  | s=12, margins [0.08,0.25], alpha=0.7 |
| Arena-10 medium | arena_tradeoff_frontier | 64  | 10  | 16    | 300  | s=16, margins [0.06,0.20], alpha=0.7 |
| Witness-4       | unique_witness_d        | 40  | 4   | 8     | 500  | s=8, q=4, margins [0.04,0.16]        |
| Witness-10      | unique_witness_d        | 80  | 10  | 16    | 300  | s=16, q=4, margins [0.035,0.14]      |
| Two-group-10    | highdim_two_group       | 64  | 10  | 15.79 | 300  | K_low=48, K_high=16                  |

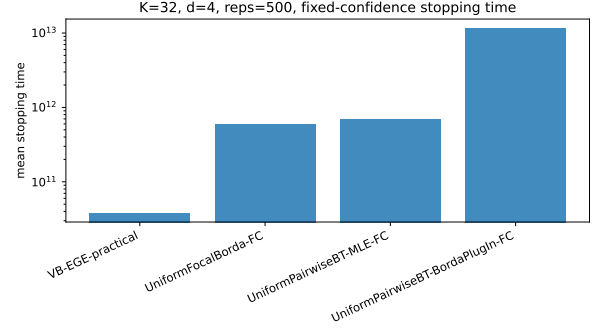
Table 7: Benchmark results. The table reports the mean stopping time of VB-EGE and the multiplicative stopping-time ratio of each baseline relative to VB-EGE. All empirical error rates are zero.

| Setting         | VB-EGE $\bar{\tau}$   | Focal/VB | MLE/VB | Plug-in/VB | Wilson upper |
|-----------------|-----------------------|----------|--------|------------|--------------|
| Convex-2D       | $1.21 \times 10^8$    | 3.55x    | 6.90x  | 118x       | 0.0076       |
| Arena-4 small   | $3.83 \times 10^{10}$ | 15.7x    | 18.3x  | 302x       | 0.0076       |
| Arena-4 medium  | $3.65 \times 10^{10}$ | 31.6x    | 69.1x  | 1227x      | 0.0126       |
| Arena-10 medium | $9.79 \times 10^8$    | 13.0x    | 27.6x  | 500x       | 0.0126       |
| Witness-4       | $2.28 \times 10^8$    | 1.25x    | 1.79x  | 29.7x      | 0.0076       |
| Witness-10      | $1.61 \times 10^9$    | 2.15x    | 2.95x  | 48.7x      | 0.0126       |
| Two-group-10    | $7.16 \times 10^{11}$ | 115x     | 270x   | 4359x      | 0.0126       |

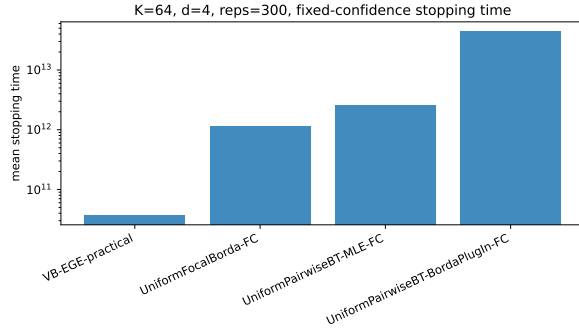
The benchmark suite stresses different Pareto geometries beyond the symmetric hard case. Convex-2D tests a clean low-dimensional trade-off frontier. Arena settings use mutually non-dominated anchors and dominated alternatives offset from a random anchor. Witness settings create dominated arms that are certified by a unique Pareto witness, which tests whether algorithms can avoid unnecessary global pairwise coverage. Two-group-10 is a high-dimensional separation task with many low-quality arms and a smaller high-quality group.



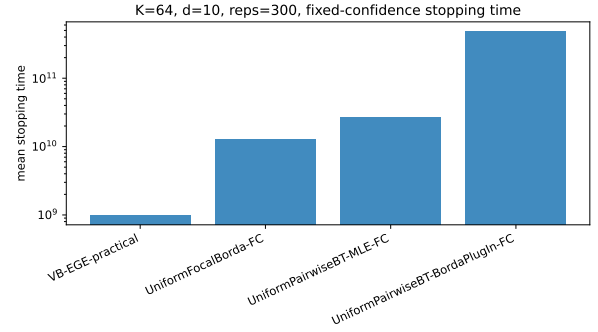
(a) Convex-2D



(b) Arena-4 small

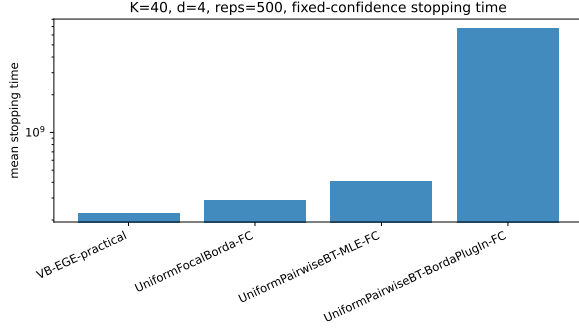


(c) Arena-4 medium

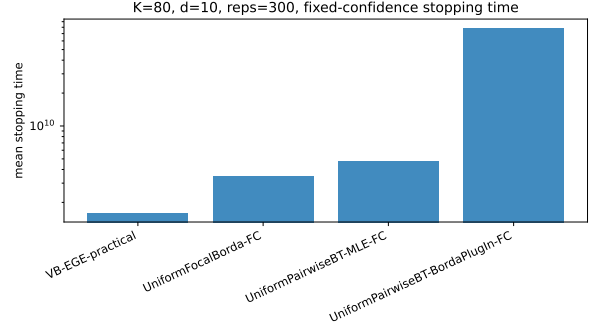


(d) Arena-10 medium

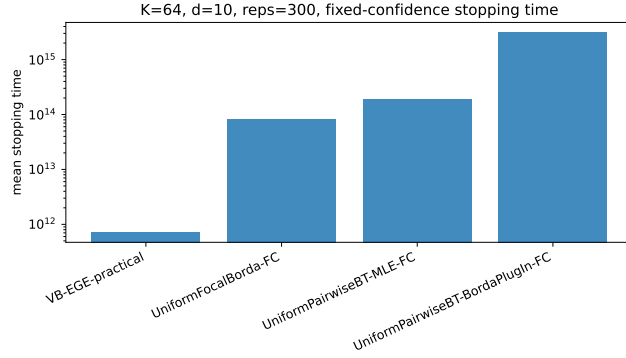
Figure 5: Benchmark stopping times, first group. The x-axis lists algorithms within each benchmark setting; the y-axis is mean stopping time  $\bar{\tau}$  on a log scale. Lower bars mean fewer comparison pulls before certification.



(a) Witness-4



(b) Witness-10



(c) Two-group-10

Figure 6: Benchmark stopping times, second group. The x-axis lists algorithms within each benchmark setting; the y-axis is mean stopping time  $\bar{\tau}$  on a log scale. These tasks emphasize local witnesses and high-dimensional group separation.

Result interpretation. VB-EGE-practical has the smallest mean stopping time in every benchmark setting while maintaining zero empirical error. The advantage is moderate on unique-witness instances, where the focal Borda baseline already receives local certificate information, and much larger on arena and two-group instances. The BT-MLE baseline has near-perfect pairwise sign accuracy in the completed runs, but it pays a large exploration cost before enough pair-coordinate cells are covered. The Borda plug-in pairwise baseline is consistently the slowest, which suggests that uniform pairwise sampling is poorly matched to fixed-confidence Pareto certification when the certificate can be expressed in the Vector-Borda embedding.

## 5 Constants Calibration

Table 8: Constant-calibration settings. VB-EGE is swept over sample constants  $\{0.5, 1, 2, 4\}$  and threshold constants  $\{2, 3, 4, 6\}$  at  $\delta = 0.05$ ; the focal and BT-MLE references use the default constants  $(2, 4)$ .

| Setting           | Generator               | Parameters                                  | Grid                                                    | Reps |
|-------------------|-------------------------|---------------------------------------------|---------------------------------------------------------|------|
| Symmetric K64 d10 | symmetric_hard          | K=64; d=10; Delta=1                         | $c_s \in \{0.5, 1, 2, 4\}; c_\theta \in \{2, 3, 4, 6\}$ | 500  |
| Arena-10 medium   | arena_tradeoff_frontier | K=64; d=10; s=16; margins [0.06,0.20]       | $c_s \in \{0.5, 1, 2, 4\}; c_\theta \in \{2, 3, 4, 6\}$ | 300  |
| Witness-10        | unique_witness_d        | K=80; d=10; s=16; q=4; margins [0.035,0.14] | $c_s \in \{0.5, 1, 2, 4\}; c_\theta \in \{2, 3, 4, 6\}$ | 300  |

Table 9: Constant-calibration results. The default row uses  $(c_s, c_\theta) = (2, 4)$ . The best row is the smallest mean stopping time among grid points with empirical error at most 0.05 and complete stopping.

| Setting           | Default VB $\bar{\tau}$ | Default err | Best constants | Best VB $\bar{\tau}$  | Focal/default | MLE/default |
|-------------------|-------------------------|-------------|----------------|-----------------------|---------------|-------------|
| Symmetric K64 d10 | $4.46 \times 10^6$      | 0.000       | (4, 2)         | $6.38 \times 10^5$    | 1.00x         | 2.27x       |
| Arena-10 medium   | $1.55 \times 10^{11}$   | 0.000       | (2, 2)         | $3.84 \times 10^{10}$ | 31.9x         | 74.9x       |
| Witness-10        | $1.62 \times 10^9$      | 0.000       | (2, 2)         | $4.75 \times 10^8$    | 2.16x         | 2.94x       |

The calibration sweep varies the VB-EGE sample constant  $c_s$  and threshold constant  $c_\theta$ . In the frontier figures, each point is one constant pair; the x-axis is mean stopping time and the y-axis is either empirical error or Wilson upper confidence bound. Heatmaps use sample constant on one axis and threshold constant on the other; darker or lighter cells encode the plotted metric named in the title. The default paper setting is  $(c_s, c_\theta) = (2, 4)$ .

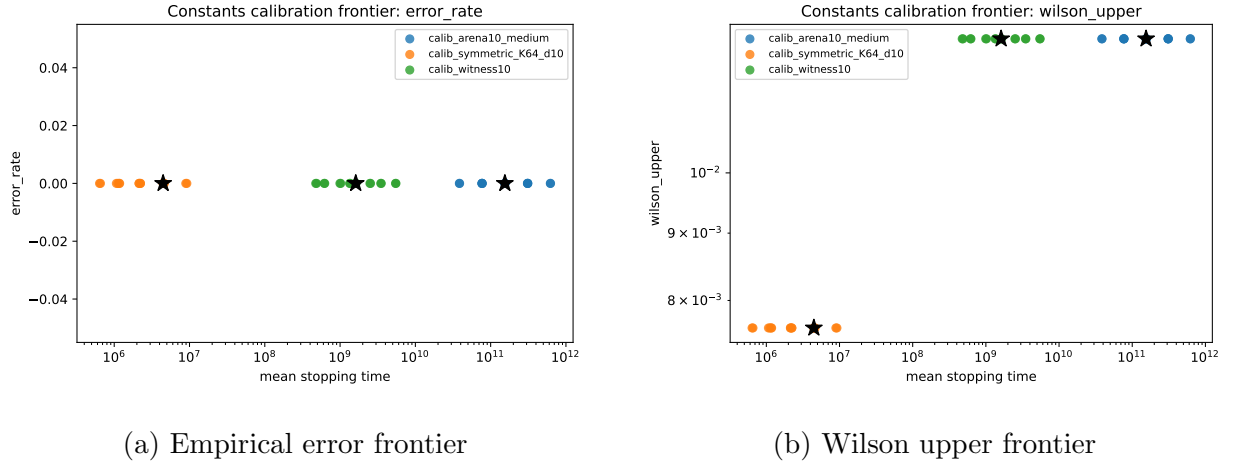
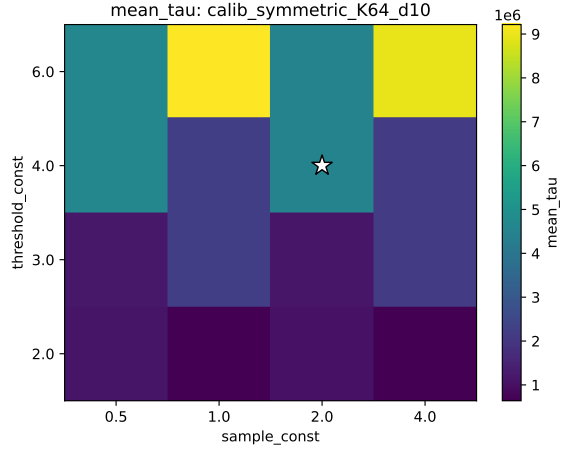
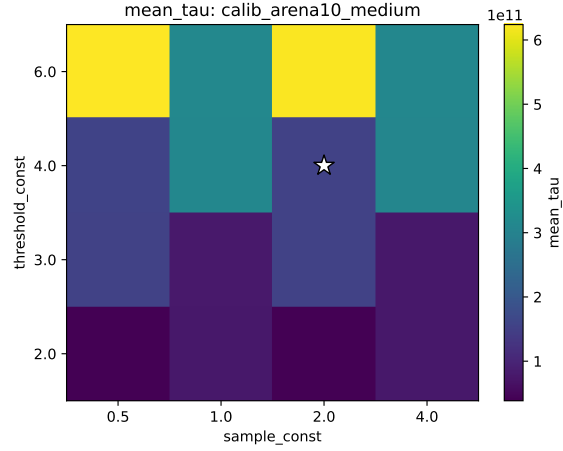


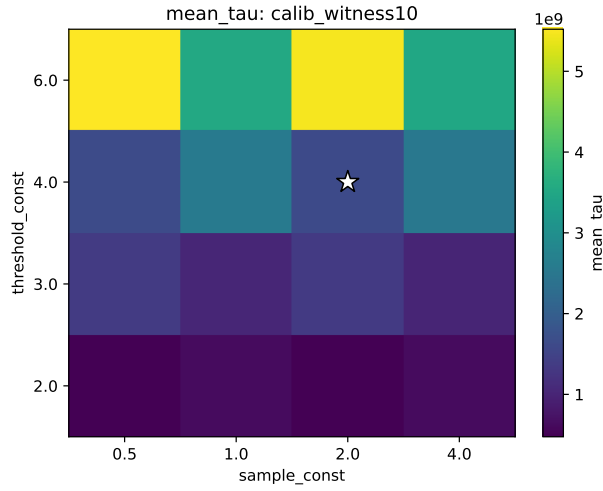
Figure 7: Calibration frontiers. The x-axis is mean stopping time  $\bar{\tau}$  for one VB-EGE constant pair. The y-axis is empirical error in panel (a) and the 95% Wilson upper error bound in panel (b). Points near the lower-left are better only if their error remains below the target confidence level.



(a) Symmetric K64 d10



(b) Arena-10 medium



(c) Witness-10

Figure 8: Calibration mean stopping-time heatmaps. The heatmap axes are sample constant  $c_s$  and threshold constant  $c_\theta$ ; each cell is one VB-EGE constant pair, and the color scale is mean stopping time  $\bar{\tau}$ .

Result interpretation. The calibration task is not intended to declare every constant pair safe; it shows the efficiency-reliability frontier. Smaller constants can reduce stopping time, but only constants with acceptable empirical error and complete stopping are viable. The default constants are therefore a conservative reference point for the main benchmark comparisons.

## 6 Pareto-Size Ablation

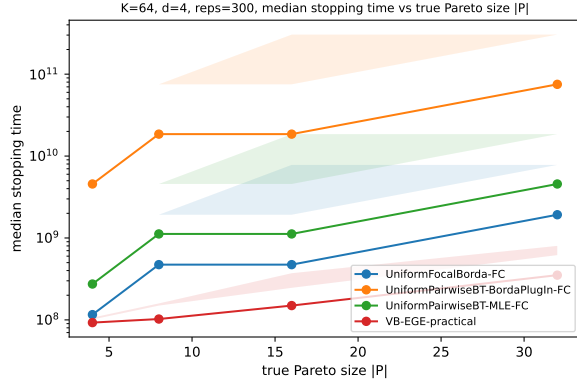
Table 10: Pareto-size ablation settings. The arena generator is held fixed while the target frontier size  $s = |P|$  changes.

| Setting  | Generator               | Grid                                              | Purpose                                                     | Reps |
|----------|-------------------------|---------------------------------------------------|-------------------------------------------------------------|------|
| Arena-4  | arena_tradeoff_frontier | K=64; d=4; s in {4,8,16,32}; margins [0.08,0.25]  | Measure how certificate cost changes as the frontier grows. | 300  |
| Arena-10 | arena_tradeoff_frontier | K=64; d=10; s in {4,8,16,32}; margins [0.06,0.20] | Measure how certificate cost changes as the frontier grows. | 300  |

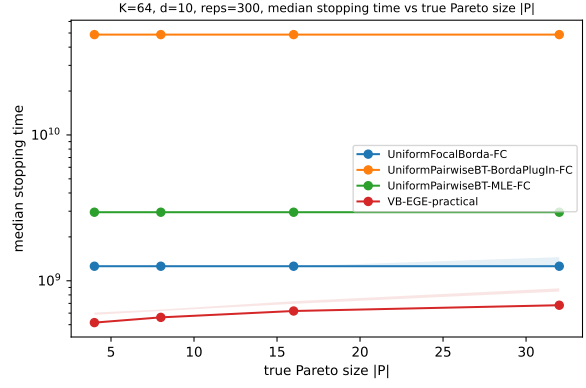
Table 11: Pareto-size ablation results. Ratios are mean stopping-time ratios relative to VB-EGE at the same arena dimension and target Pareto size.

| Setting  | $ P $ | VB $\bar{\tau}$    | VB $\tau/(dH_B)$ | Focal/VB | MLE/VB | Plug-in/VB |
|----------|-------|--------------------|------------------|----------|--------|------------|
| Arena-4  | 4     | $9.44 \times 10^7$ | 242.7            | 2.60x    | 6.11x  | 99.1x      |
| Arena-4  | 8     | $1.16 \times 10^8$ | 240.6            | 10.00x   | 22.9x  | 387x       |
| Arena-4  | 16    | $2.52 \times 10^8$ | 245.5            | 21.8x    | 39.7x  | 640x       |
| Arena-4  | 32    | $4.42 \times 10^8$ | 249.0            | 11.8x    | 28.2x  | 462x       |
| Arena-10 | 4     | $5.21 \times 10^8$ | 253.4            | 2.41x    | 5.66x  | 93.5x      |
| Arena-10 | 8     | $5.55 \times 10^8$ | 264.2            | 2.27x    | 5.32x  | 87.9x      |
| Arena-10 | 16    | $6.10 \times 10^8$ | 279.0            | 2.13x    | 4.99x  | 82.4x      |
| Arena-10 | 32    | $6.88 \times 10^8$ | 285.6            | 2.19x    | 5.09x  | 83.7x      |

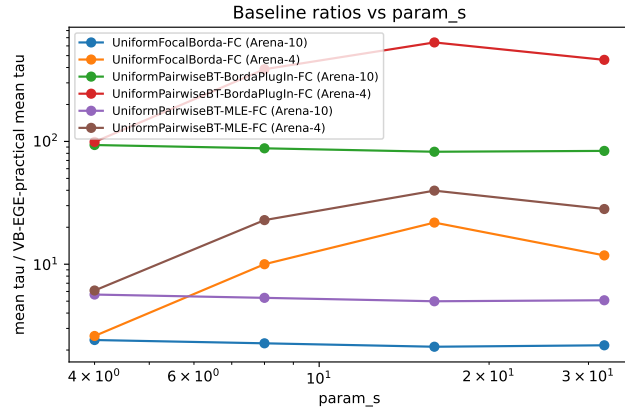
This ablation holds the arena geometry fixed while increasing the true Pareto frontier size  $|P|$ . In the plots, x-axis values are the target frontier sizes and each legend entry is an algorithm. The baseline-ratio plot reports mean stopping time divided by VB-EGE at the same setting; values above one favor VB-EGE.



(a) Arena-4 tau



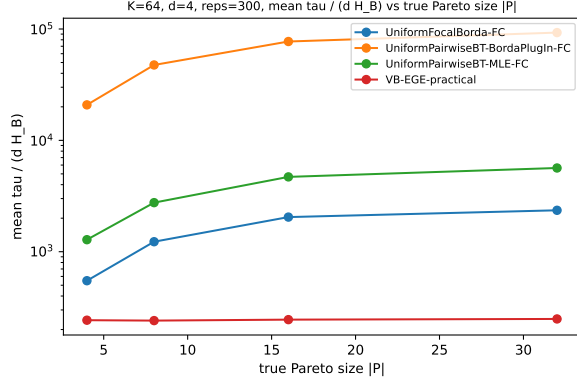
(b) Arena-10 tau



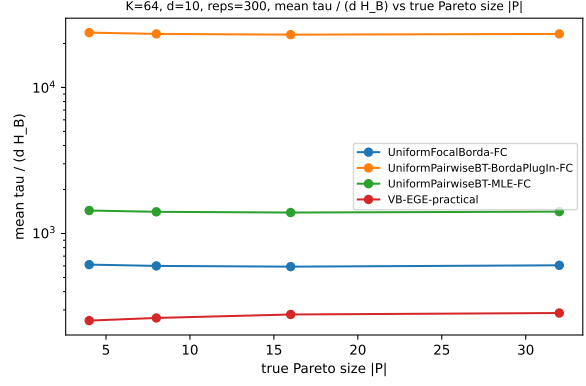
(c) Baseline/VB ratio

Figure 9: Pareto-size stopping-time ablation. The x-axis is target Pareto size  $|P|$ . Tau panels use mean stopping time  $\bar{\tau}$  on the y-axis; the ratio panel uses  $\bar{\tau}_{\text{baseline}}/\bar{\tau}_{\text{VB}}$ .





(a) Arena-4 normalized



(b) Arena-10 normalized

Figure 10: Pareto-size normalized stopping times. The x-axis is target Pareto size  $|P|$  and the y-axis is mean  $\tau/(dH_B)$ . Normalization by  $dH_B$  separates hardness changes from pure frontier-size effects.

Result interpretation. The important question is whether the relative advantage of VB-EGE persists as the Pareto frontier grows. When the ratio curves stay above one, the adaptive Vector-Borda certificate remains more sample-efficient than uniform baselines even though more arms need to be accepted.

## 7 Correlated Arena

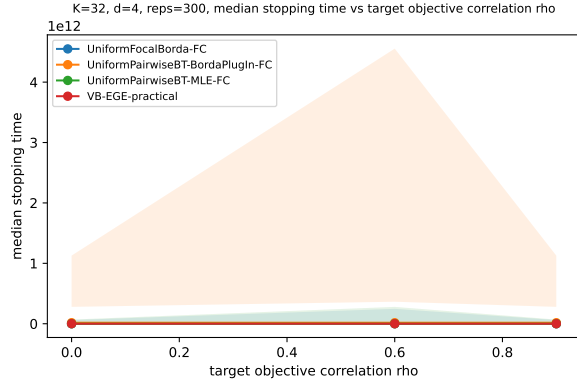
Table 12: Correlated-arena settings. The correlated generator constructs frontier anchors from shared latent factors and records the achieved average off-diagonal objective correlation.

| Setting             | Generator             | Grid                                           | Purpose                                                      | Reps |
|---------------------|-----------------------|------------------------------------------------|--------------------------------------------------------------|------|
| Correlated Arena-4  | correlated_arena_like | K=32; d=4; s=8; rho in {0,0.6,0.9}             | Test whether correlated objectives change sample complexity. | 300  |
| Correlated Arena-10 | correlated_arena_like | K=64; d=10; s=16; rho in {0,0.3,0.6,0.85,0.95} | Test whether correlated objectives change sample complexity. | 300  |

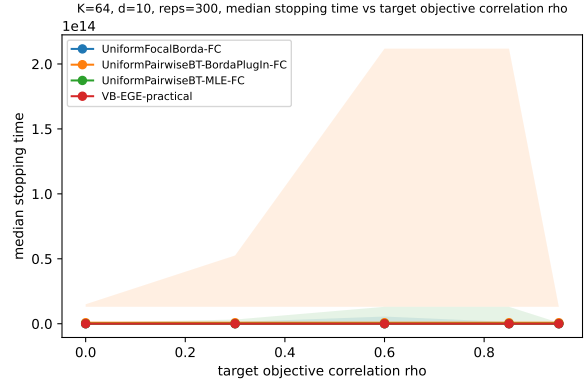
Table 13: Correlated-arena results. Achieved corr. is the mean off-diagonal objective correlation produced by the generator.

| Setting             | Target $\rho$ | Achieved corr. | VB $\bar{\tau}$       | VB $\tau/(dH_B)$ | Focal/VB | MLE/VB | Plug-in/VB |
|---------------------|---------------|----------------|-----------------------|------------------|----------|--------|------------|
| Correlated Arena-4  | 0             | -0.09          | $6.16 \times 10^{10}$ | 232.2            | 15.5x    | 18.2x  | 296x       |
| Correlated Arena-4  | 0.60          | -0.04          | $6.42 \times 10^{10}$ | 237.9            | 16.0x    | 18.1x  | 293x       |
| Correlated Arena-4  | 0.90          | -0.03          | $2.05 \times 10^9$    | 238.9            | 14.7x    | 16.2x  | 272x       |
| Correlated Arena-10 | 0             | 0.06           | $1.46 \times 10^{10}$ | 258.6            | 30.7x    | 69.2x  | 1171x      |
| Correlated Arena-10 | 0.30          | 0.16           | $1.03 \times 10^{13}$ | 257.8            | 32.0x    | 75.1x  | 1213x      |
| Correlated Arena-10 | 0.60          | 0.22           | $2.54 \times 10^{11}$ | 273.8            | 28.3x    | 74.2x  | 1211x      |
| Correlated Arena-10 | 0.85          | 0.25           | $4.94 \times 10^{11}$ | 268.5            | 53.7x    | 57.3x  | 2062x      |
| Correlated Arena-10 | 0.95          | 0.25           | $1.01 \times 10^{10}$ | 274.1            | 54.2x    | 68.0x  | 1164x      |

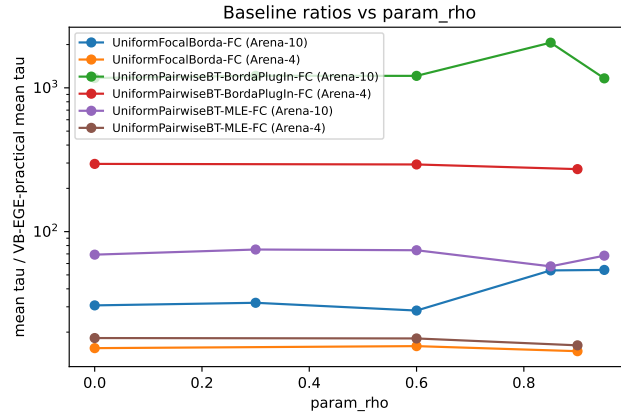
The correlated arena generator attempts to vary a target objective-correlation parameter  $\rho$ , but the achieved correlations are modest because the generator also enforces a requested Pareto-front size. Therefore these runs should be read as diagnostics rather than a strong correlation-robustness claim. The achieved-correlation column reports the actual mean off-diagonal correlation after resampling. Curves are grouped by target  $\rho$  only for bookkeeping; achieved correlation is the descriptive quantity to inspect. The pair-cell coverage diagnostic is meaningful only for pairwise baselines, while the Borda-gap plot reports the average minimum Borda certificate gap.



(a) Arena-4 tau

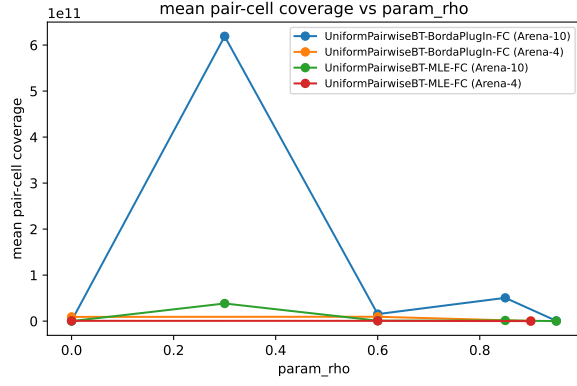


(b) Arena-10 tau

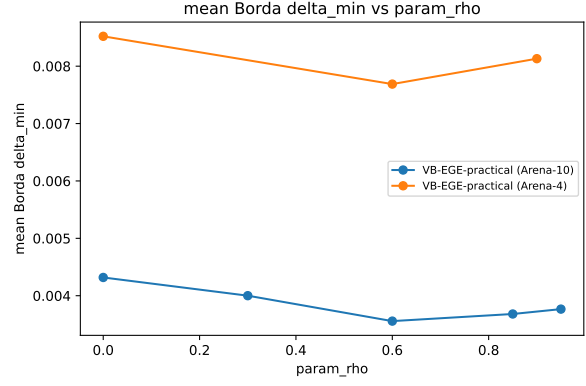


(c) Baseline/VB ratio

Figure 11: Correlated-arena stopping times. The x-axis is target  $\rho$ . Tau panels use mean stopping time  $\bar{\tau}$ ; the ratio panel uses  $\bar{\tau}_{\text{baseline}}/\bar{\tau}_{\text{VB}}$ . Lower tau curves and higher ratios favor VB-EGE.



(a) Pair-cell coverage

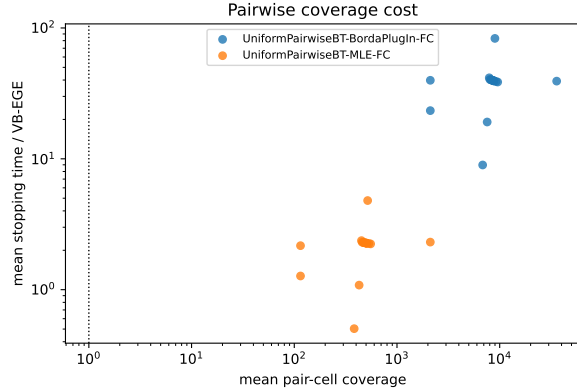


(b) Minimum Borda gap

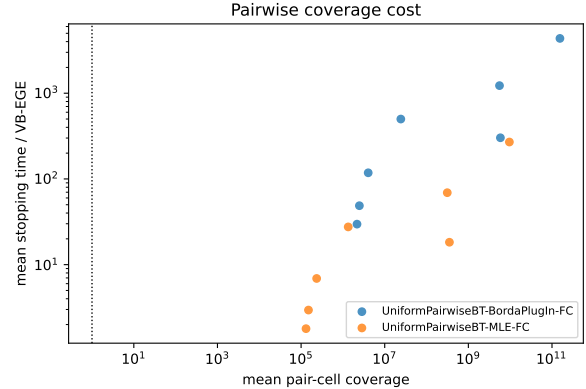
Figure 12: Correlated-arena diagnostics. The x-axis is target  $\rho$ . The coverage y-axis is mean pair-cell coverage  $\tau/C$  for pairwise baselines, with  $C = dK(K-1)/2$ ; the gap y-axis is mean  $\Delta_{\min,B}$ .

Result interpretation. Correlation changes both the geometry of the objectives and the effective Borda gaps. If stopping time increases while the minimum Borda gap shrinks, the harder certificate explains the cost. If pairwise baselines grow faster than VB-EGE at similar Borda gaps, the extra cost comes from their broad direct pair-coordinate sampling requirement.

## 8 Pair-Cell Coverage Diagnostics



(a) Scaling tasks



(b) Benchmark tasks

Figure 13: Pair-cell coverage cost diagnostics. Only pairwise baselines appear. The x-axis is mean pair-cell coverage  $\tau/C$ , where  $C = dK(K-1)/2$ ; the y-axis is  $\bar{\tau}_{\text{baseline}}/\bar{\tau}_{\text{VB}}$  on the same setting.

The coverage diagnostic explains a major source of the stopping-time gap. Each point is one pairwise baseline on one setting; higher coverage and higher ratio mean that the method used broader direct pair-coordinate evidence and took more samples than VB-EGE on the same task. VB-EGE instead

estimates coordinate-wise Vector-Borda scores and can stop after certifying accepted and rejected arms in that lower-dimensional certificate space.

## 9 Fixed-Error Budget Calibration

Table 14: Fixed-error budget calibration settings. Budgets are reported as fractions of the default fixed-confidence VB-EGE mean stopping time on the matching benchmark task.

| Setting         | Generator               | Parameters                                  | Budget ratios                    | $\tau_{VB,ref}$ | Reps |
|-----------------|-------------------------|---------------------------------------------|----------------------------------|-----------------|------|
| Arena-10 medium | arena_tradeoff_frontier | K=64; d=10; s=16; margins [0.06,0.20]       | {0.02, 0.05, 0.1, 0.2, 0.5, 1.0} | 9.79e8          | 300  |
| Witness-10      | unique_witness_d        | K=80; d=10; s=16; q=4; margins [0.035,0.14] | {0.02, 0.05, 0.1, 0.2, 0.5, 1.0} | 1.61e9          | 300  |

Table 15: Fixed-error budget ratios. Entries are the first tested budget ratio  $T/\bar{\tau}_{VB,ref}$  with empirical error at most 5%. Ratios compare each baseline budget ratio with VB-EGE-capped on the same setting.

| Setting         | VB-capped | Focal/VB | MLE/VB | Plug-in/VB |
|-----------------|-----------|----------|--------|------------|
| Arena-10 medium | 0.02      | 1.00x    | 1.00x  | 1.00x      |
| Witness-10      | 0.10      | 0.50x    | 0.20x  | 0.20x      |

This calibration is not the baseline definition. The fixed-confidence baseline algorithms above remain the main comparison. Here, capped variants are run under a fixed budget to show what happens before the certification rule has enough samples. The x-axis is the budget divided by the default fixed-confidence VB-EGE mean stopping time, and the table reports the first tested budget ratio reaching empirical error at most 5%.

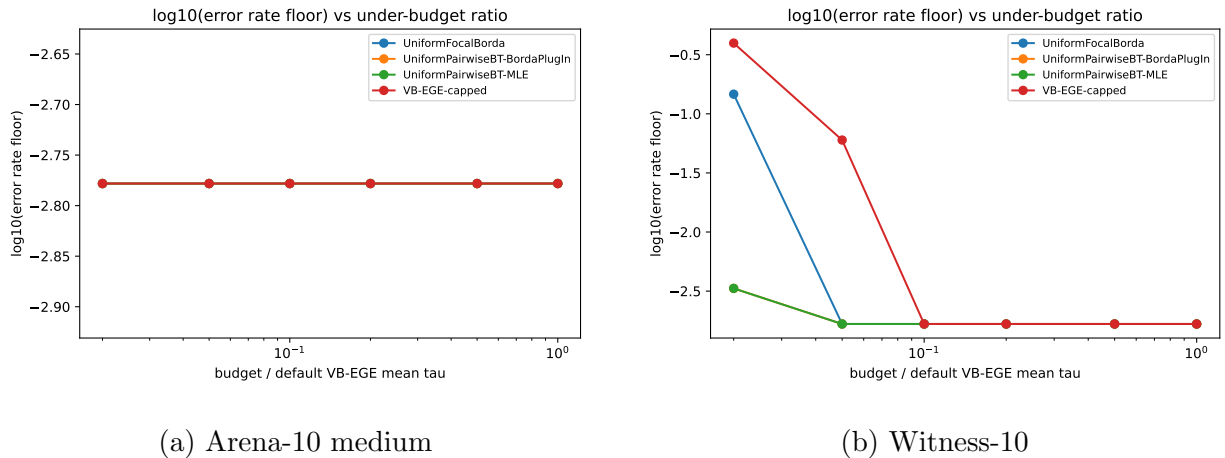
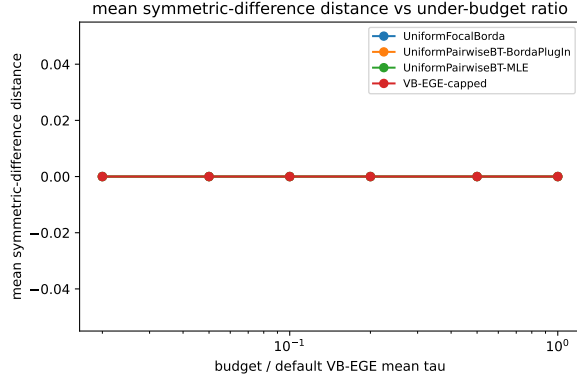
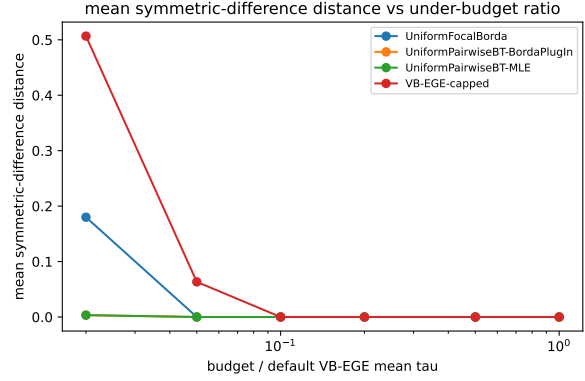


Figure 14: Fixed-error budget calibration: empirical error. The x-axis is budget ratio  $T/\bar{\tau}_{VB,ref}$  and the y-axis is  $\log_{10}$  empirical error, with the finite-sample floor used only when a method has no observed failures.



(a) Arena-10 medium

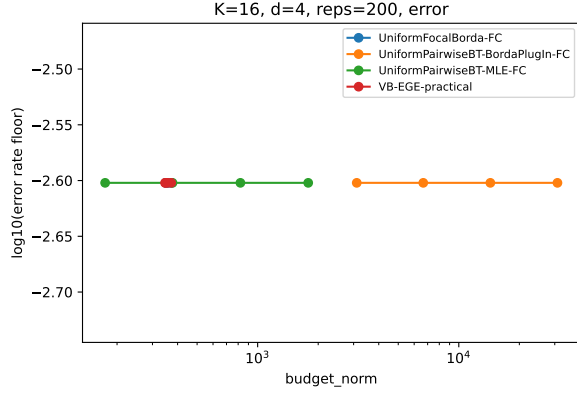


(b) Witness-10

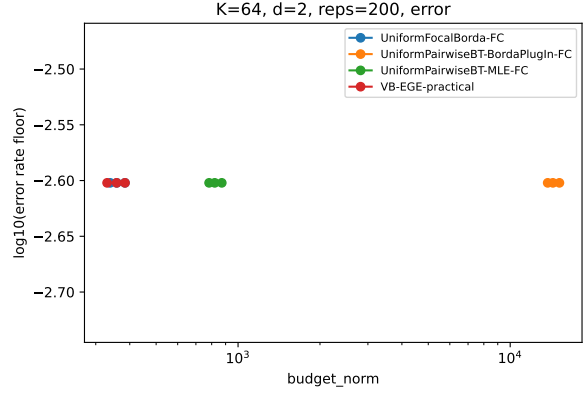
Figure 15: Fixed-error budget calibration: Hamming distance. The x-axis is budget ratio  $T/\bar{\tau}_{\text{VB,ref}}$  and the y-axis is mean symmetric-difference distance  $|\hat{P} \Delta P|$ .

## 10 Zero-Error Diagnostics

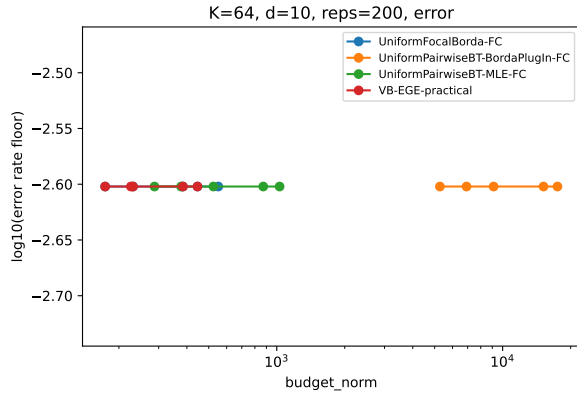
These plots diagnose the fixed-confidence runs whose empirical error is zero. Since the observed empirical error is zero throughout the main fixed-confidence sweeps, their y-values are  $\log_{10}(0.5/n)$  floors rather than measured error rates.



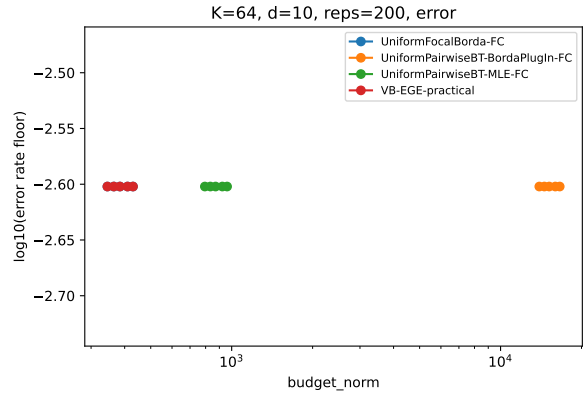
(a) K scaling



(b) d scaling

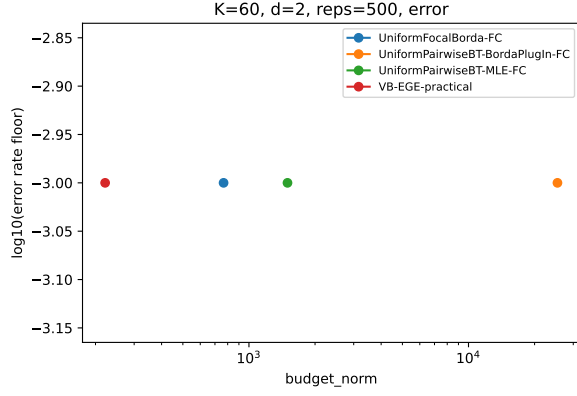


(c) Gap scaling

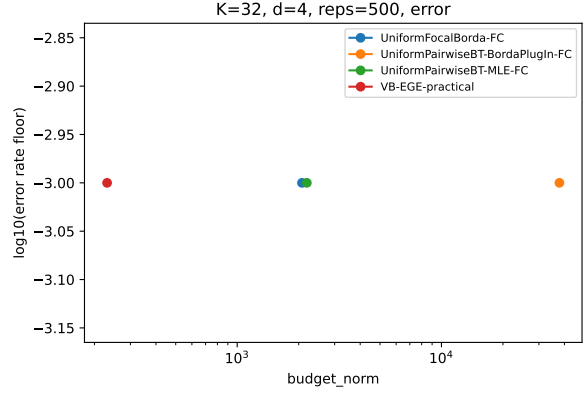


(d) Confidence scaling

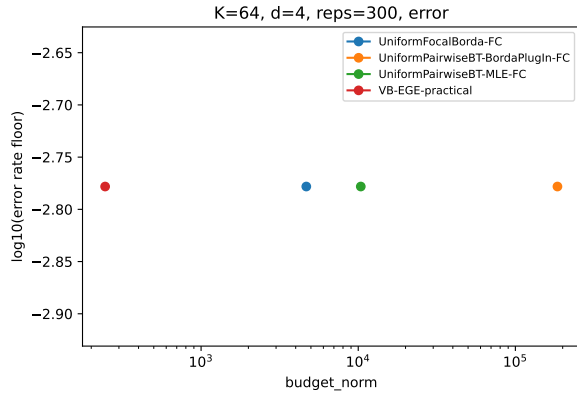
Figure 16: Scaling error-rate floors under normalized budget. The x-axis is normalized budget  $\tau/(dH_B)$  and the y-axis is  $\log_{10}(0.5/n)$  because all empirical errors are zero.



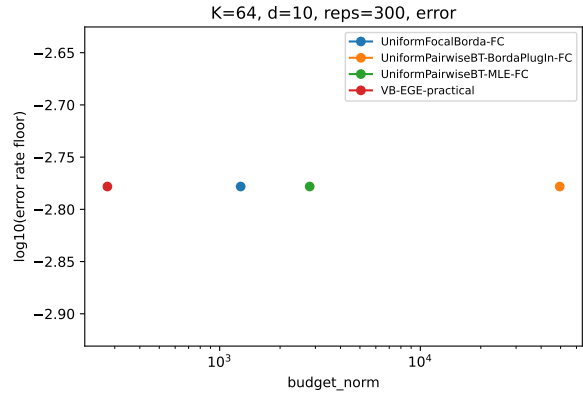
(a) Convex-2D



(b) Arena-4 small



(c) Arena-4 medium



(d) Arena-10 medium

Figure 17: Benchmark normalized error floors, first group. The x-axis is normalized budget  $\tau/(dH_B)$  and the y-axis is  $\log_{10}(0.5/n)$  because all empirical errors are zero.



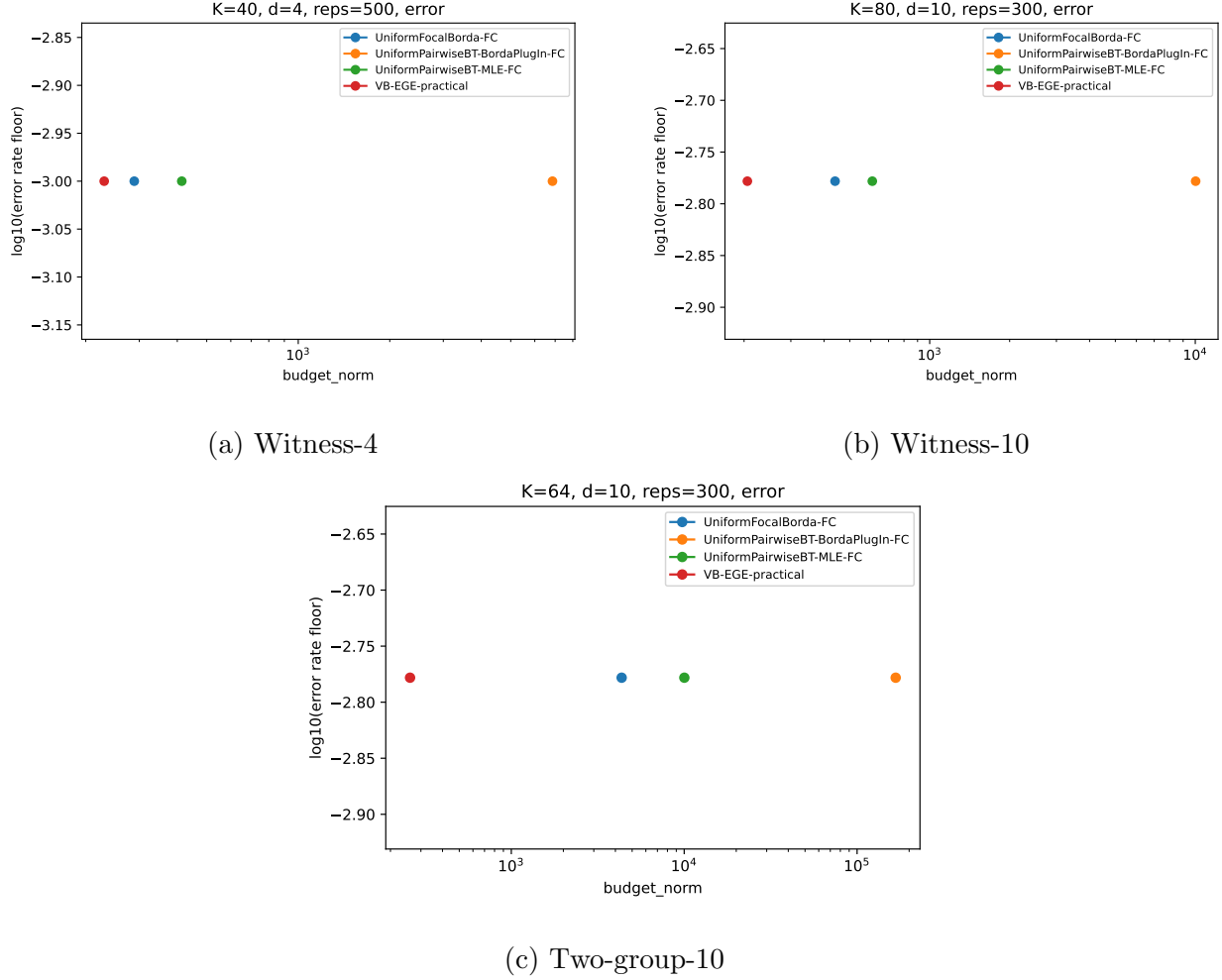


Figure 18: Benchmark normalized error floors, second group. The x-axis is normalized budget  $\tau/(dH_B)$  and the y-axis is  $\log_{10}(0.5/n)$  because all empirical errors are zero. Curves at the same floor indicate exact empirical recovery across all replications.

## 11 Takeaways

1. All completed main fixed-confidence synthetic experiments achieved zero empirical Pareto-set error.
2. VB-EGE-practical is the most sample-efficient method on every benchmark setting and is competitive with the focal Borda baseline on the symmetric hard scaling task.
3. Pairwise BT-MLE can estimate latent pairwise signs accurately, but fixed-confidence Pareto certification makes its broad pair-coordinate coverage expensive.
4. The largest gains occur in high-dimensional arena and two-group settings, where adaptive Vector-Borda certificates avoid many unnecessary pairwise comparisons.

5. The reported baselines are the fixed-confidence baseline algorithms; the fixed-error budget calibration is a separate pre-certification stress test, not a replacement baseline.

## Reproducibility Notes

The source summaries used in this report are `results/summary/fixed_confidence_scaling_summary.csv` and `results/summary/fixed_confidence_benchmarks_summary.csv`, plus `results/summary/confidence_scaling_quantile_summary.csv`, `results/summary/constants_calibration_summary.csv`, `results/summary/pareto_size_ablation_summary.csv`, and `results/summary/correlated_arena_summary.csv`, plus `results/summary/under_budget_stress_summary.csv` for the fixed-error budget calibration. The raw completed main outputs are `results/raw/fixed_confidence_scaling.csv`, `results/raw/fixed_confidence_benchmarks.csv`, and the corresponding CSV fallbacks under `results/raw/` for the diagnostic extensions. Smoke runs are excluded from the main conclusions because they were used only to validate the implementation path before the full runs.